ECEN 356: Probability and Random Variables
Spring 2026

Assignment 1: Due date February 27, 2026 (via blackboard Everything in one PDF file)
File name: ECEN-356-2026-Spring-LastName-FirstName-Asst-1
Write your answers with a clear handwriting/type
(I suggest you to use latex or word equation editor to avoid confusion when needed)
When the question/part of the question requires a numerical value for final answer, make sure that the final answer is correctly calculated

Feedback No.	COMMENT	Percentage value
F1	Did the student provide an explanation on the problem (Identify given and required variables)?	15.00%
F2	Did the student clearly show the steps to solve the problem?	25.00%
F3	Did the student use the correct formula?	25.00%
F4	Did the student correctly replace the variables in the formula?	20.00%
F5	Did the student find the correct answer?	15.00%

Problem 1

Suppose that the universal set S is defined as $S = \{1, 2, \cdots, 10\}$ and $A = \{1, 2, 3\}$, $B = \{X \in S : 2 \leq X \leq 7\}$, and $C = \{7, 8, 9, 10\}$.

a. Find $A \cup B$.
b. Find $(A \cup C) - B$.
c. Find $\bar{A} \cup (B - C)$.
d. Do A, B, and C form a partition of S?

Problem 13

Two teams A and B play a soccer match, and we are interested in the winner. The sample space can be defined as

$$S = \{a, b, d\},$$

where a shows the outcome that A wins, b shows the outcome that B wins, and d shows the outcome that they draw. Suppose we know that:
(1) the probability that A wins is $P(a) = P(\{a\}) = 0.5$;
(2) the probability of a draw is $P(d) = P(\{d\}) = 0.25$.

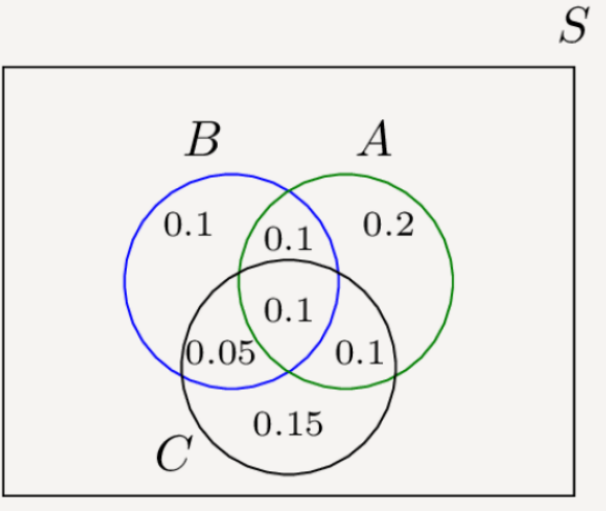
a. Find the probability that B wins.
b. Find the probability that B wins or a draw occurs.

Problem 17

Four teams A, B, C, and D compete in a tournament, and exactly one of them will win the tournament. Teams A and B have the same chance of winning the tournament. Team C is twice as likely to win the tournament as team D. The probability that either team A or team C wins the tournament is 0.6. Find the probabilities of each team winning the tournament.

Problem 23

Let $A, B,$ and C be three events with probabilities given below:



a. Find $P(A|B)$.
b. Find $P(C|B)$.
c. Find $P(B|A \cup C)$.
d. Find $P(B|A, C) = P(B|A \cap C)$.

Problem 25

A professor thinks students who live on campus are more likely to get As in the probability course. To check this theory, the professor combines the data from the past few years:

a. 600 students have taken the course,
b. 120 students have gotten As,
c. 200 students lived on campus,
d. 80 students lived off campus and got As.

Does this data suggest that "getting an A" and "living on campus" are dependent or independent?

① a) $\boxed{A \cup B = \{1,2,3,4,5,6,7\}}$

b) $(A \cup C) - B$ $\quad A \cup C = \{1,2,3,7,8,9,10\}$

$\boxed{(A \cup C) - B = \{1,8,9,10\}}$

c) $A \cup (B - C)$ $\quad B - C = \{2,3,4,5,6\}$

$\boxed{A \cup (B - C) = \{1,2,3,4,5,6\}}$

d) No, A, B, C do not form a partition because they overlap

⑬ $S = \{a, b, d\}$ $\quad P(a) = 0.50$ $\quad P(d) = 0.25$

a) $P(a,b,c) = 1 \rightarrow P(b) = P(a,b,c) - P(a) - P(d)$

$= 1 - 0.50 - 0.25$

$\boxed{P(b) = 0.25}$

b) $P(b \cup d) = P(b) + P(d)$

$= 0.25 + 0.25$

$\boxed{P(b \cup d) = 0.50}$

⑰ $P(A) = P(B)$ $\quad P(C) = 2P(D)$ $\quad P(A \cup C) = 0.60$

$P(A) = P(B) = x$ $\quad P(D) = y$ $\quad P(C) = 2y$

$P(A) + P(B) + P(C) + P(D) = 1 \rightarrow x + x + 2y + y = 1$

$2x + 3y = 1$

$P(A \cup C) = P(A) + P(C) \rightarrow x + 2y = 0.60$

$$\left.\begin{aligned} 2x + 3y &= 1 \\ x + 2y &= 0.60 \end{aligned}\right\} \quad x = 0.20 \quad y = 0.20$$

$$\boxed{P(A) = P(B) = 0.20 \quad P(D) = 0.20 \quad P(C) = 0.40}$$

23

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(A) = 0.5 \quad P(B) = 0.35 \quad P(C) = 0.4$

$P(A \cap B) = 0.2 \quad P(A \cap C) = 0.2$

$P(B \cap C) = P(C \cap B) = 0.15$

$P(A \cap B \cap C) = 0.1$

a) $P(A|B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.2}{0.35} = 0.57$

b) $P(C|B) = \dfrac{P(C \cap B)}{P(B)} = \dfrac{0.15}{0.35} = 0.43$

c) $P(B|A \cup C) = \dfrac{P[B \cap (A \cup C)]}{P(A \cup C)} = \dfrac{P[(B \cap A) \cup (B \cap C)]}{P(A \cup C)}$

$P(A \cup C) = 0.1 + 0.2 + 0.15 + 0.05 + 0.1 + 0.1 = 0.7$

$P(B|A \cup C) = \dfrac{0.2 + 0.15 - 0.1}{0.7} = 0.357$

d) $P(B|A\cap C) = \dfrac{P[B\cap(A\cap C)]}{A\cap C} = \dfrac{P(A\cap B\cap C)}{A\cap C}$

$P(B|A\cap C) = \dfrac{0.1}{0.2} = 0.5$

25 $\Sigma S = 600$ # of S w/ A = 120 # S on Campus = 200

S off Campus = 400 off Campus S w/ A = 80

ON Campus S w/ A = 120 − 80 = 40

$P(A \mid ON) = \dfrac{\text{ON Campus S w/ A}}{\text{\# of S on Campus}} = \dfrac{40}{200} = 0.2$

$P(A \mid OFF) = \dfrac{\text{off Campus S w/ A}}{\text{\# S off Campus}} = \dfrac{80}{400} = 0.2$

$P(A) = \dfrac{\text{\# of S w/ A}}{\Sigma S} = \dfrac{120}{600} = 0.2$

$\boxed{P(A \mid ON) = P(A \mid OFF)\text{, so living on Campus is independent}}$